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MIL-M-28787/537(SH)
20 October 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RS 232C INTERFACE ADAPTER MODULE, KEY CODE AML

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic RS 232C interface adapter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.749 failures per million operating hours for class I or predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AML.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A35	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power supply stress	IV	S	--------Module shall pass all remaining tests----------						
D1,D2	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1,E2	Ios	VI	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
E3-E6	Ios	VI	S	-100.0	-20.0	-100.0	-20.0	-100.0	-20.0	mA
F1-F7	Iih	VII	X	-1.6	0.04	-1.6	0.04	-1.6	0.04	mA
F8	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F9	Iih	VII	X	-1.6	0.04	-1.6	0.04	-1.6	0.04	mA
F10	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G1-G7	Iil	VIII	X	-1.6	0.04	-1.6	0.04	-1.6	0.04	mA
G8	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G9	Iil	VIII	X	-1.6	0.04	-1.6	0.04	-1.6	0.04	mA
G10	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
H1-H6	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J6	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1,L2	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XII	S	1.4	2.6	1.4	2.6	1.4	2.6	uF
M2	PSFC	XII	S	3.0	5.6	3.0	5.6	3.0	5.6	uF
M3	PSFC	XII	S	0.9	1.7	0.9	1.7	0.9	1.7	uF
M4	PSFC	XII	S	1.5	2.7	1.5	2.7	1.5	2.7	uF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
N1	Icc	XIII	X	150.0	360.0	150.0	360.0	150.0	360.0	mA
N2	Icc	XIII	X	17.0	41.0	17.0	41.0	17.0	41.0	mA
N3	Icc	XIII	X	-9.4	-3.9	-9.4	-3.9	-9.4	-3.9	mA
P1,P2	Functional	XIV	X	----------See table XIV----------						
Q1,Q2	Dynamic	XV	S	----------See table XV----------						
Q3,Q4	Dynamic	XV	Q	----------See table XV----------						
R1-R4	Cin	XVII	Q	4	50	4	50	4	50	pF
R5	Cin	XVII	Q	2	75	2	75	2	75	pF
R6,R7	Cin	XVII	Q	4	75	4	75	4	75	pF
R8	Cin	XVII	Q	4	50	4	50	4	50	pF
R9-R11	Cin	XVII	Q	4	75	4	75	4	75	pF
R12	Cin	XVII	Q	4	100	4	100	4	100	pF
R13-R15	Cin	XVII	Q	2	50	2	50	2	50	pF
R16	Cin	XVII	Q	4	75	4	75	4	75	pF
R17-R19	Cin	XVII	Q	2	50	2	50	2	50	pF
R20,R21	Cin	XVII	Q	2	75	2	75	2	75	pF
R22,R23	Cin	XVII	Q	4	50	4	50	4	50	pF
R24	Cin	XVII	Q	2	75	2	75	2	75	pF
T1-T4	PSTR	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/537-1 for class I and M28787/537-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc, (V2) of 16.5 V dc, and (V3) of -16.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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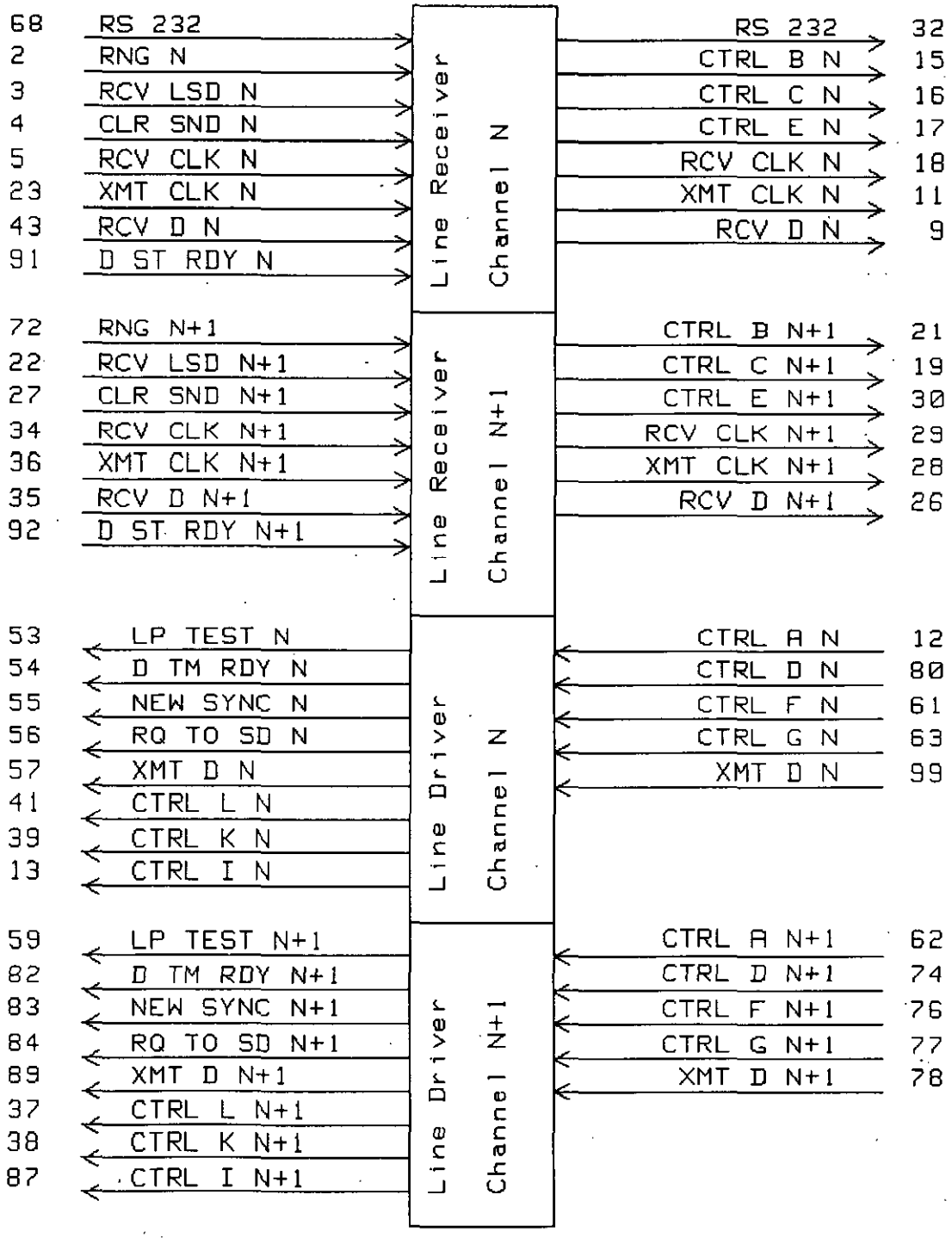


FIGURE 1. AML functional block diagram.

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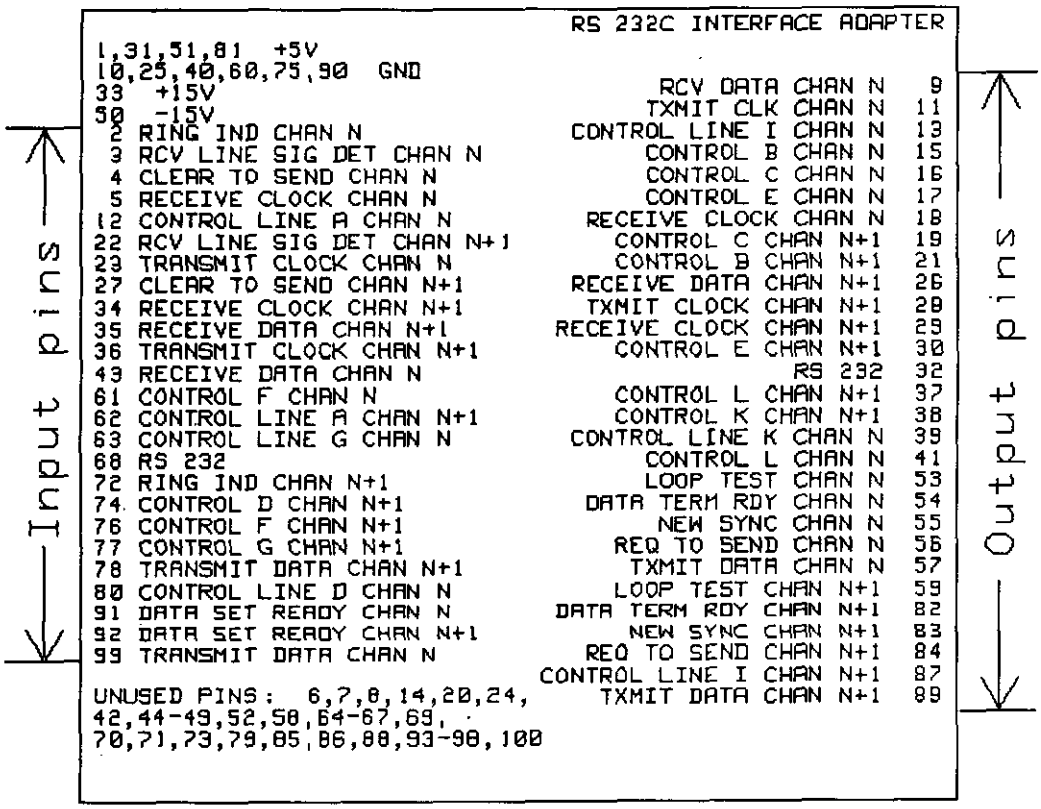


FIGURE 2. AML pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

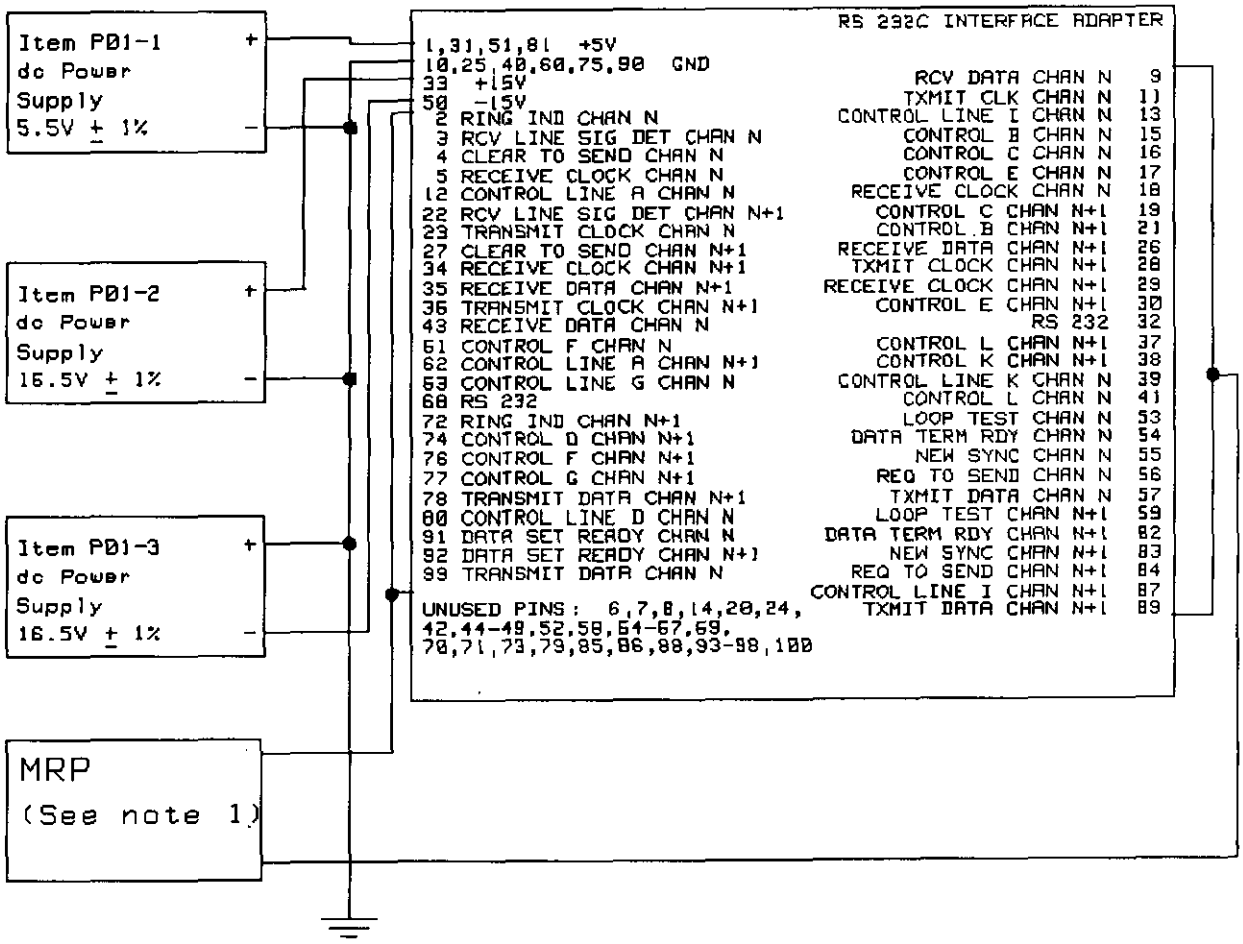
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AML operating temperature and component temperature test circuit.

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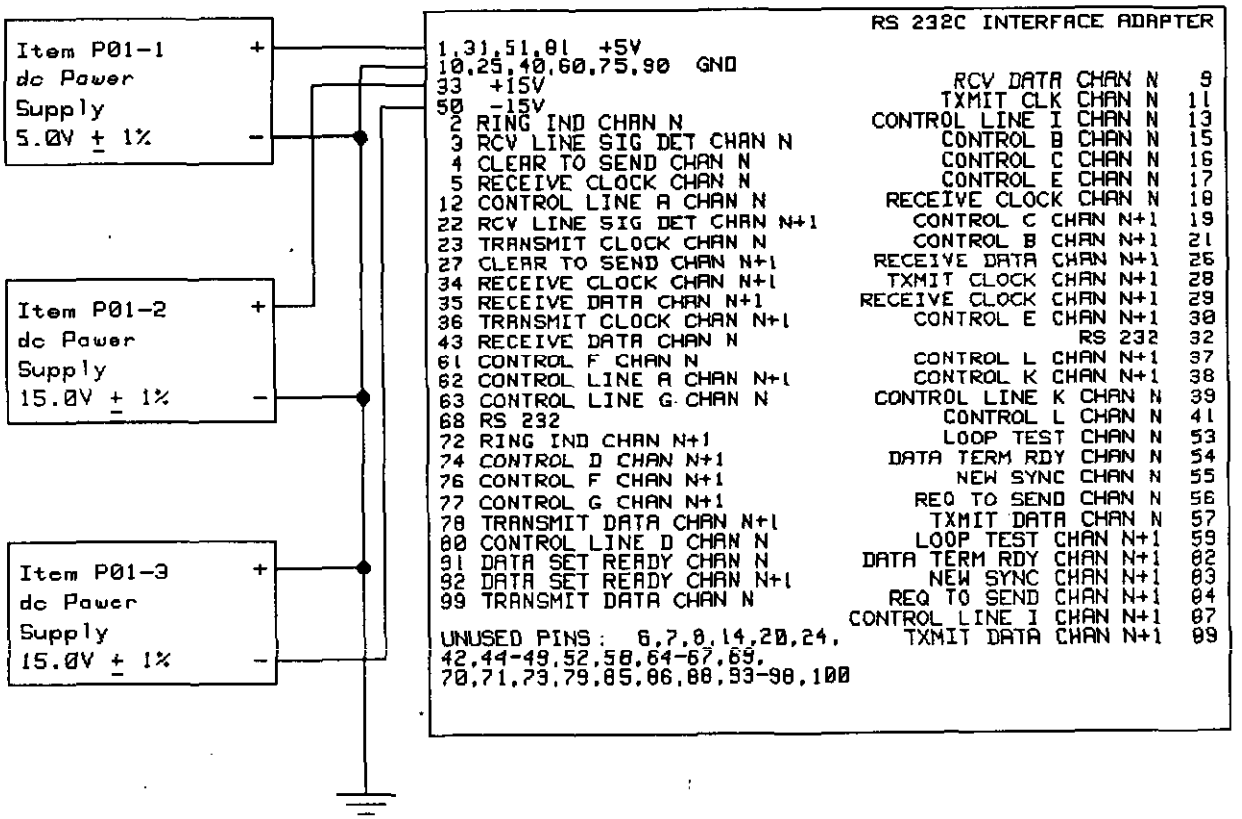


FIGURE 4. AML vibration test circuit.

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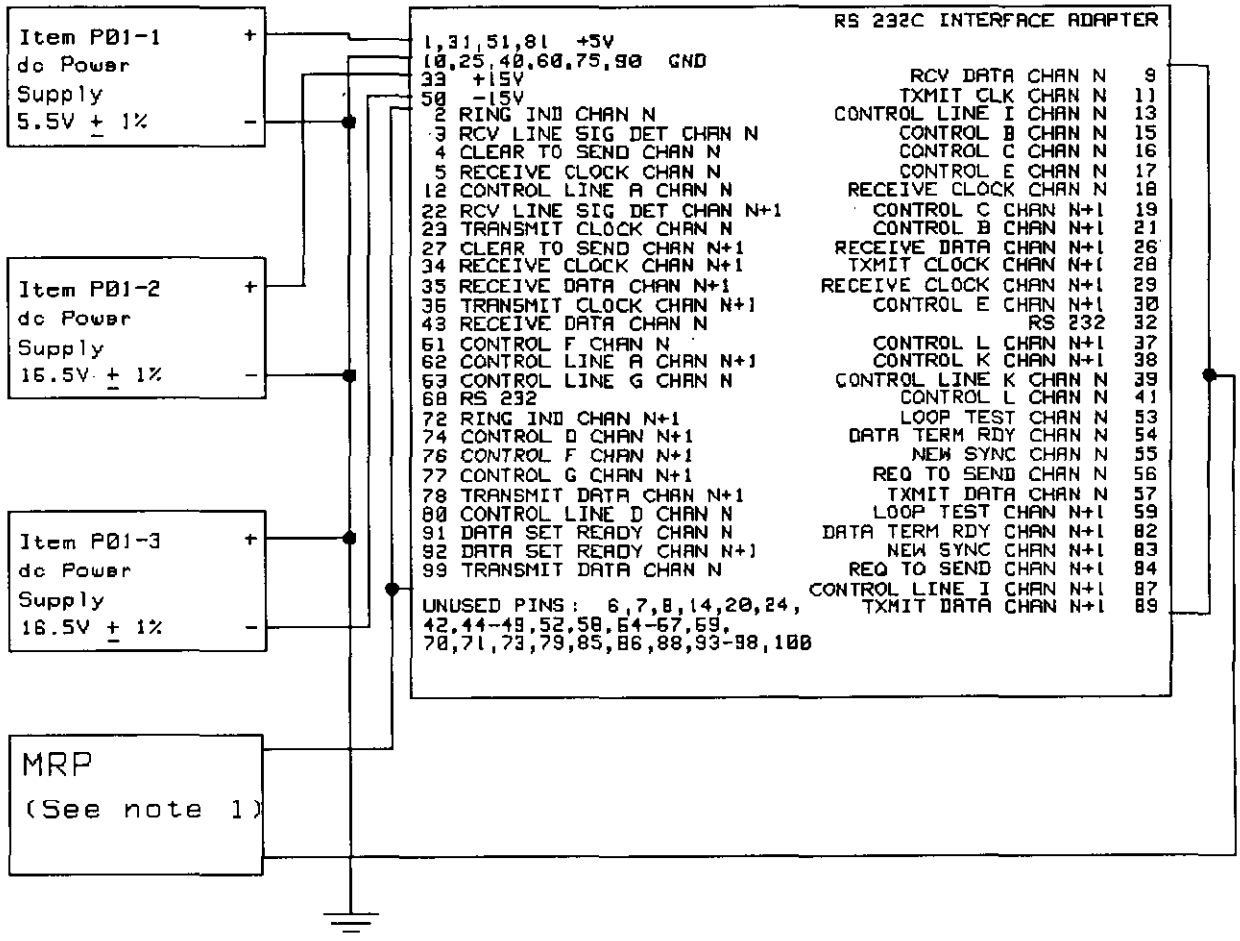
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AML life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) NA = not applicable.
(k) PSFC = power supply filter capacitance.
(l) PSTA = power supply transient amplitude.
(m) Vic = input clamp diode voltage.
(n) Vih = input high level voltage.
(o) Vil = input low level voltage.
(p) Vin = input forcing voltage.
(q) Voh = output high level voltage.
(r) Vol = output low level voltage.
(s) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 4.5 V dc to 5.5 V dc, (V2) 13.5 V dc to 16.5 V dc, and (V3) -16.5 V dc to -13.5 V dc. Module power dissipation: 2.81 W (max); 1.88 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RNG N	NA	-4.0	4.0	NA	NA	NA	50
RCV LSD N	NA	-4.0	4.0	NA	NA	NA	50
CLR SND N	NA	-4.0	4.0	NA	NA	NA	50
RCV CLK N	NA	-4.0	4.0	NA	NA	NA	50
CTRL A N	NA	0.4	2.4	-1.6	0.04	NA	75
RCV LSD N+1	NA	-4.0	4.0	NA	NA	NA	75
XMT CLK N	NA	-4.0	4.0	NA	NA	NA	75
CLR SND N+1	NA	-4.0	4.0	NA	NA	NA	50
RCV CLK N+1	NA	-4.0	4.0	NA	NA	NA	75
RCV D N+1	NA	-4.0	4.0	NA	NA	NA	75
XMT CLK N+1	NA	-4.0	4.0	NA	NA	NA	75
RCV D N	NA	-4.0	4.0	NA	NA	NA	100
CTRL F N	NA	0.4	2.4	-1.6	0.04	NA	50
CTRL A N+1	NA	0.4	2.4	-1.6	0.04	NA	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CTRL G N	NA	0.4	2.4	-1.6	0.04	NA	50
RNG N+1	NA	-4.0	4.0	NA	NA	NA	75
CTRL D N+1	NA	0.4	2.4	-1.6	0.04	NA	50
CTRL F N+1	NA	0.4	2.4	-1.6	0.04	NA	50
CTRL G N+1	NA	0.4	2.4	-1.6	0.04	NA	50
XMT D N+1	-1.5	0.4	2.4	-0.8	0.04	0.1	75
CTRL D N	NA	0.4	2.4	-1.6	0.04	NA	75
D ST RDY N	NA	NA	NA	NA	NA	NA	50
D ST RDY N+1	NA	NA	NA	NA	NA	NA	50
XMT D N	-1.5	0.4	2.4	-0.8	0.04	0.1	75
RS 232	NA	0.4	2.4	NA	NA	NA	NA

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)
RCV D N	2.4	0.5	-0.16	1.6	-15.0
RCV D N+1	2.4	0.5	-0.16	1.6	-15.0
CTRL L N+1	2.4	NA	-0.16	NA	-20.0
CTRL K N+1	2.4	0.5	-0.16	1.6	-20.0
CTRL K N	2.4	0.5	-0.16	1.6	-20.0
CTRL L N	2.4	NA	-0.16	NA	-20.0
CTRL I N	NA	0.5	NA	1.6	NA
CTRL I N+1	NA	0.5	NA	1.6	NA
LP TEST N	6.0	-6.0	-0.4	6.0	NA
D TM RDY N	6.0	-6.0	-0.4	6.0	NA
NEW SYNC N	6.0	-6.0	-0.4	6.0	NA
RQ TO SD N	6.0	-6.0	-0.4	6.0	NA
XMT D N	6.0	-6.0	-0.4	6.0	NA
LP TEST N+1	6.0	-6.0	-0.4	6.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)
D TM RDY N+1	6.0	-6.0	-0.4	6.0	NA
NEW SYNC N+1	6.0	-6.0	-0.4	6.0	NA
RQ TO SD N+1	6.0	-6.0	-0.4	6.0	NA
XMT D N+1	6.0	-6.0	-0.4	6.0	NA
XMT CLK N	2.4	0.5	-0.4	6.0	NA
CTRL B N	2.4	0.5	-0.4	6.0	NA
CTRL C N	2.4	0.5	-0.4	6.0	NA
CTRL E N	2.4	0.5	-0.4	6.0	NA
RCV CLK N	2.4	0.5	-0.4	6.0	NA
CTRL C N+1	2.4	0.5	-0.4	6.0	NA
CTRL B N+1	2.4	0.5	-0.4	6.0	NA
XMT CLK N+1	2.4	0.5	-0.4	6.0	NA
RCV CLK N+1	2.4	0.5	-0.4	6.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)
CTRL E N+1	2.4	0.5	-0.4	6.0	NA
RS 232	NA	NA	NA	NA	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	RNG N to CTRL B N	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV LSD N to CTRL C N	-	-	-	3500	3500	3500	nS	11
1	tpd	CLR SND N to CTRL E N	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV CLK N to RCV CLK N	-	-	-	3500	3500	3500	nS	11
1	tpd	XMT CLK N to XMT CLK N	-	-	-	3500	3500	3500	nS	11
1	tpd	RNG N+1 to CTRL B N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV LSD N+1 to CTRL C N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	CLR SND N+1 to CTRL E N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV CLK N+1 to RCV CLK N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	XMT CLK N+1 to XMT CLK N+1	-	-	-	3500	3500	3500	nS	11
2	tpd	RCV D N to RCV D N	-	-	-	3527	3527	3527	nS	11
2	tpd	RCV D N+1 to RCV D N+1	-	-	-	3527	3527	3527	nS	11
3	tpd	D ST RDY N to CTRL K N	-	-	-	3545	3545	3545	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	D ST RDY N+1 to CTRL K N+1	-	-	-	3545	3545	3545	nS	11
4	tpd	CTRL A N to LP TEST N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL D N to D TM RDY N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL F N to NEW SYNC N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL G N to RQ TO SD N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL A N+1 to LP TEST N+1	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL D N+1 to D TM RDY N+1	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL F N+1 to NEW SYNC N+1	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL G N+1 to RQ TO SD N+1	-	-	-	4500	4500	4500	nS	11
5	tpd	XMT D N+1 to XMT D N+1	-	-	-	4527	4527	4527	nS	11
5	tpd	XMT D N to XMT D N	-	-	-	4527	4527	4572	nS	11

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6.4 Module function. The RS 232C serial interface module (AML) contains line drivers and line receivers for serial channels N and N+1.

6.4.1 Line drivers. The driver circuits for the interface lines accept the Transistor Transistor Logic (TTL) level signal and convert it to RS 232C levels for transmission to the external device. Inhibit lines to the drivers are placed at signal ground to ensure a constant enable. Driver outputs are short-circuit protected and fault protected. Output slew rate (rise time and fall time) for each driver is externally limited by a capacitor placed in parallel with the output signal. Supply voltages for the drivers (+ 12 V) are on-module regulated from the + 15 V supply. Transmit data signals destined for channel N and N+1 are exclusive OR'ed with the RS 232 signal (ground) to provide proper data polarity for the RS 232C interface.

6.4.2 Line receivers. Each receiver circuit for the interface lines accept RS 232C voltage levels and converts it to TTL level. The receivers are used in the differential mode of operation to allow discrimination between the driving signal and noise. Additionally, the receivers employ internal fail-safe circuitry that keeps outputs high when inputs are open. Receive data for channel N and N+1 is exclusive OR'ed with the RS 232 signal to provide proper data polarity for the RS 232C interface.

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6.4.3 Description of I/O pins.

Signal name	Pin number	I/O	Description
RNG N	2	I	RING IND CHAN N
RCV LSD N	3	I	RCV LINE SIG DET CHAN N
CLR SND N	4	I	CLEAR TO SEND CHAN N
RCV CLK N	5	I	RECEIVE CLOCK CHAN N
CTRL A N	12	I	CONTROL LINE A CHAN N
RCV LSD N+1	22	I	RCV LINE SIG DET CHAN N+1
XMT CLK N	23	I	TRANSMIT CLOCK CHAN N
CLR SND N+1	27	I	CLEAR TO SEND CHAN N+1
RCV CLK N+1	34	I	RECEIVE CLOCK CHAN N+1
RCV D N+1	35	I	RECEIVE DATA CHAN N+1
XMT CLK N+1	36	I	TRANSMIT CLOCK CHAN N+1
RCV D N	43	I	RECEIVE DATA CHAN N
CTRL F N	61	I	CONTROL F CHAN N
CTRL A N+1	62	I	CONTROL LINE A CHAN N+1
CTRL G N	63	I	CONTROL LINE G CHAN N
RS 232	68	I	RS 232
RNG N+1	72	I	RING IND CHAN N+1
CTRL D N+1	74	I	CONTROL D CHAN N+1
CTRL F N+1	76	I	CONTROL F CHAN N+1
CTRL G N+1	77	I	CONTROL G CHAN N+1

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
XMT D N+1	78	I	TRANSMIT DATA CHAN N+1
CTRL D N	80	I	CONTROL LINE D CHAN N
D ST RDY N	91	I	DATA SET READY CHAN N
D ST RDY N+1	92	I	DATA SET READY CHAN N+1
XMT D N	99	I	TRANSMIT DATA CHAN N
RCV D N	9	O	RCV DATA CHAN N
XMT CLK N	11	O	TXMIT CLK CHAN N
CTRL I N	13	O	CONTROL LINE I CHAN N
CTRL B N	15	O	CONTROL B CHAN N
CTRL C N	16	O	CONTROL C CHAN N
CTRL E N	17	O	CONTROL E CHAN N
RCV CLK N	18	O	RECEIVE CLOCK CHAN N
CTRL C N+1	19	O	CONTROL C CHAN N+1
CTRL B N+1	21	O	CONTROL B CHAN N+1
RCV D N+1	26	O	RECEIVE DATA CHAN N+1
XMT CLK N+1	28	O	TXMIT CLOCK CHAN N+1
RCV CLK N+1	29	O	RECEIVE CLOCK CHAN N+1
CTRL E N+1	30	O	CONTROL E CHAN N+1
RS 232	32	O	RS 232
CTRL L N+1	37	O	CONTROL L CHAN N+1

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CTRL K N+1	38	O	CONTROL K CHAN N+1
CTRL K N	39	O	CONTROL LINE K CHAN N
CTRL L N	41	O	CONTROL L CHAN N
LP TEST N	53	O	LOOP TEST CHAN N
D TM RDY N	54	O	DATA TERM RDY CHAN N
NEW SYNC N	55	O	NEW SYNC CHAN N
RQ TO SD N	56	O	REQ TO SEND CHAN N
XMT D N	57	O	TXMIT DATA CHAN N
LP TEST N+1	59	O	LOOP TEST CHAN N+1
D TM RDY N+1	82	O	DATA TERM RDY CHAN N+1
NEW SYNC N+1	83	O	NEW SYNC CHAN N+1
RQ TO SD N+1	84	O	REQ TO SEND CHAN N+1
CTRL I N+1	87	O	CONTROL LINE I CHAN N+1
XMT D N+1	89	O	TXMIT DATA CHAN N+1

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N731)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,6,7,8,14,20,24,42,44,45,46,47,48,49,52,58,64,65,66,67,69,70,71,73,79,85,86,88,93,94,95,96,97,98,100	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	6,7,8,14,20,24,42,44,45,46,47,48,49,52,58,64,65,66,67,69,70,71,73,79,85,86,88,93,94,95,96,97,98,100				
A3-A35	Isolation, unused to unused pins	See note 4.		10	INF	MOhm	1-9

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A35
Isolation measurement	NA	± 500 KOhm	A1-A35

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 6,7,8,14,20,24,42,44,45, 46,47,48,49,52,58,64,65,66,67,69,70,71,73,79,85,86,88,93,94,95, 96,97,98,100. When testing for unused to unused pin isolation connect one unused pin to A and the remaining unused pins to B.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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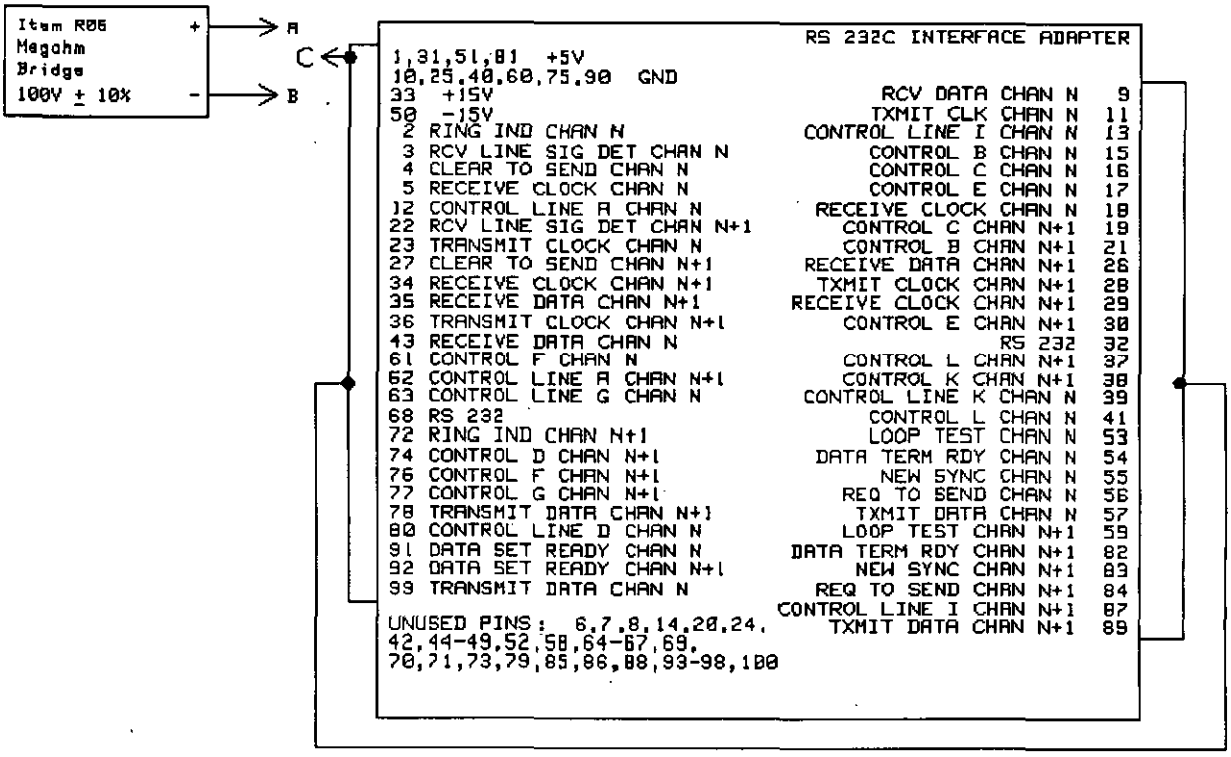


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

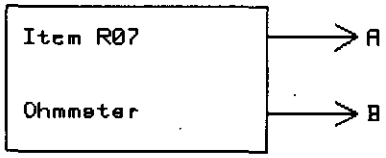
2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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RS 232C INTERFACE ADAPTER

1,31,51,81	+5V		
10,25,40,60,75,90	GND		
33	+15V	RCV DATA CHAN N	9
50	-15V	TXMIT CLK CHAN N	11
2	RING IND CHAN N	CONTROL LINE I CHAN N	13
3	RCV LINE SIG DET CHAN N	CONTROL B CHAN N	15
4	CLEAR TO SEND CHAN N	CONTROL C CHAN N	16
5	RECEIVE CLOCK CHAN N	CONTROL E CHAN N	17
12	CONTROL LINE A CHAN N	RECEIVE CLOCK CHAN N	18
22	RCV LINE SIG DET CHAN N+1	CONTROL C CHAN N+1	19
23	TRANSMIT CLOCK CHAN N	CONTROL B CHAN N+1	21
27	CLEAR TO SEND CHAN N+1	RECEIVE DATA CHAN N+1	26
34	RECEIVE CLOCK CHAN N+1	TXMIT CLOCK CHAN N+1	28
35	RECEIVE DATA CHAN N+1	RECEIVE CLOCK CHAN N+1	29
36	TRANSMIT CLOCK CHAN N+1	CONTROL E CHAN N+1	30
43	RECEIVE DATA CHAN N	RS 232	32
61	CONTROL F CHAN N	CONTROL L CHAN N+1	37
62	CONTROL LINE A CHAN N+1	CONTROL K CHAN N+1	38
63	CONTROL LINE G CHAN N	CONTROL LINE K CHAN N	39
68	RS 232	CONTROL L CHAN N	41
72	RING IND CHAN N+1	LOOP TEST CHAN N	53
74	CONTROL D CHAN N+1	DATA TERM RDY CHAN N	54
76	CONTROL F CHAN N+1	NEW SYNC CHAN N	55
77	CONTROL G CHAN N+1	REQ TO SEND CHAN N	56
78	TRANSMIT DATA CHAN N+1	TXMIT DATA CHAN N	57
80	CONTROL LINE D CHAN N	LOOP TEST CHAN N+1	59
91	DATA SET READY CHAN N	DATA TERM RDY CHAN N+1	82
92	DATA SET READY CHAN N+1	NEW SYNC CHAN N+1	83
99	TRANSMIT DATA CHAN N	REQ TO SEND CHAN N+1	84
		CONTROL LINE I CHAN N+1	87
UNUSED PINS:	6,7,8,14,20,24,	TXMIT DATA CHAN N+1	89
42,44-49,52,58,64-67,69,			
70,71,73,79,85,86,88,93-98,100			

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
C	V2	17.5	33	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
C	V3	17.5	10,25, 40,60, 75,90	50	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	7.0 V	+0 mV,-70 mV	C
(V2)	17.5 V	+0 mV,-175 mV	C
(V3)	-17.5 V	+175 mV,-0 mV	C
Power supply current limit,			
(Vcc)	0.65 A	NA	C
(V2)	0.08 A	NA	C
(V3)	-0.02 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

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NOTES: - Continued.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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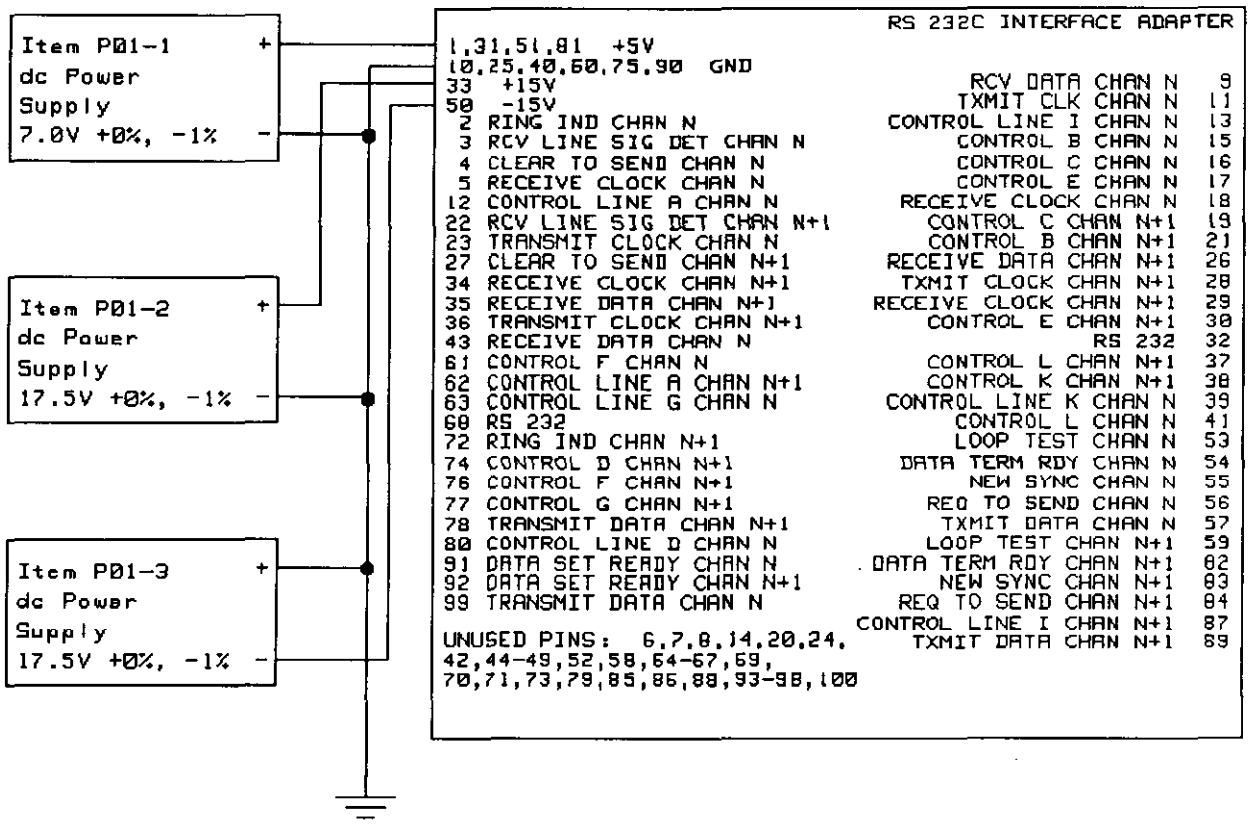


FIGURE 8. Power supply stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	XMT D N+1	78	5.5	0.0	0.1	mA	1-3
D2	XMT D N	99	5.5	0.0	0.1	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1,D2
(V2)	16.5 V	± 165 mV	D1,D2
(V3)	-16.5 V	± 165 mV	D1,D2
Power supply current limit, (Vcc)	0.54 A	NA	D1,D2
(V2)	0.06 A	NA	D1,D2
(V3)	-0.02 A	NA	D1,D2
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1,D2
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1,D2
low, (VL)	0.0 V	± 100 mV	D1,D2
Input current measurement	NA	± 5 uA	D1,D2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	RCV D N	9	0.0	-100.0	-15.0	mA	1-4
E2	RCV D N+1	26	0.0	-100.0	-15.0	mA	1-4
E3	CTRL L N+1	37	0.0	-100.0	-20.0	mA	1-4
E4	CTRL K N+1	38					
E5	CTRL K N	39					
E6	CTRL L N	41	0.0	-100.0	-20.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E6
(V2)	16.5 V	± 165 mV	E1-E6
(V3)	-16.5 V	± 165 mV	E1-E6
Power supply current limit, (Vcc)	0.54 A	NA	E1-E6
(V2)	0.06 A	NA	E1-E6
(V3)	-0.02 A	NA	E1-E6
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E6

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E6 E1-E6
Output current measurement	 NA NA	 ± 4.25 mA ± 4.00 mA	 E1,E2 E3-E6

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	CTRL A N	12	2.4	-1.6	0.04	mA	1-3
F2	CTRL F N	61					
F3	CTRL A N+1	62					
F4	CTRL G N	63					
F5	CTRL D N+1	74					
F6	CTRL F N+1	76					
F7	CTRL G N+1	77	2.4	-1.6	0.04	mA	1-3
F8	XMT D N+1	78	2.4	-0.8	0.04	mA	1-3
F9	CTRL D N	80	2.4	-1.6	0.04	mA	1-3
F10	XMT D N	99	2.4	-0.8	0.04	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F10
(V2)	16.5 V	± 165 mV	F1-F10
(V3)	-16.5 V	± 165 mV	F1-F10
Power supply current limit, (Vcc)	0.54 A	NA	F1-F10
(V2)	0.06 A	NA	F1-F10
(V3)	-0.02 A	NA	F1-F10

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F10
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F10 F1-F10
Input current measurement	 NA NA NA NA	 ± 82 uA ± 42 uA ± 82 uA ± 42 uA	 F1-F7 F8 F9 F10

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CTRL A N	12	0.4	-1.6	0.04	mA	1-3
G2	CTRL F N	61					
G3	CTRL A N+1	62					
G4	CTRL G N	63					
G5	CTRL D N+1	74					
G6	CTRL F N+1	76					
G7	CTRL G N+1	77	0.4	-1.6	0.04	mA	1-3
G8	XMT D N+1	78	0.4	-0.8	0.04	mA	1-3
G9	CTRL D N	80	0.4	-1.6	0.04	mA	1-3
G10	XMT D N	99	0.4	-0.8	0.04	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G10
(V2)	16.5 V	± 165 mV	G1-G10
(V3)	-16.5 V	± 165 mV	G1-G10
Power supply current limit, (Vcc)	0.54 A	NA	G1-G10
(V2)	0.06 A	NA	G1-G10
(V3)	-0.02 A	NA	G1-G10

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G10
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G10 G1-G10
Input current measurement	NA NA NA NA	± 82 uA ± 42 uA ± 82 uA ± 42 uA	G1-G7 G8 G9 G10

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	RCV D N	9	-0.16	2.4	4.5	V	1-4
H2	RCV D N+1	26					
H3	CTRL L N+1	37					
H4	CTRL K N+1	38					
H5	CTRL K N	39					
H6	CTRL L N	41	-0.16	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H6
(V2)	13.5 V	± 135 mV	H1-H6
(V3)	-13.5 V	± 135 mV	H1-H6
Power supply current limit, (Vcc)	0.54 A	NA	H1-H6
(V2)	0.06 A	NA	H1-H6
(V3)	-0.02 A	NA	H1-H6
Forcing current, (Ioh)	-0.16 A	± 16 uA	H1-H6

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NOTES: - Continued

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H6 H1-H6
Output voltage measurement	NA	± 105 mV	H1-H6

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

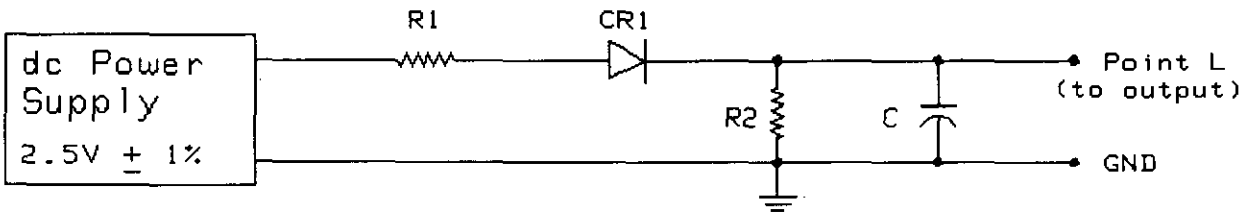
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
11,15,16,17,18, 19,21,28,29,30	6200	6.0	0.5
53,54,55,56,57, 59,82,83,84,89	6200	6.0	-6.0
9,26,38,39	16000	1.6	0.5
37,41	16000	NA	NA
13,87	NA	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	RCV D N	9	1.6	0.0	0.5	V	1-4
J2	CTRL I N	13					
J3	RCV D N+1	26					
J4	CTRL K N+1	38					
J5	CTRL K N	39					
J6	CTRL I N+1	87	1.6	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J6
(V2)	13.5 V	± 135 mV	J1-J6
(V3)	-13.5 V	± 135 mV	J1-J6
Power supply current limit, (Vcc)	0.54 A	NA	J1-J6
(V2)	0.06 A	NA	J1-J6
(V3)	-0.02 A	NA	J1-J6
Forcing current, (Iol)	1.6 mA	± 160 uA	J1-J6

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J6 J1-J6
Output voltage measurement	NA	± 25 mV	J1-J6

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	XMT D N+1	78	-18.0	-1.5	0.0	V	1-3
L2	XMT D N	99	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2) (V3)	Off Off Off	NA NA NA	L1,L2 L1,L2 L1,L2
Power supply current limit, (Vcc) (V2) (V3)	NA NA NA	NA NA NA	L1,L2 L1,L2 L1,L2
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1,L2
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	L1,L2 L1,L2
Input voltage measurement	NA	± 75 mV	L1,L2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC, Vcc	1	1.4	2.6	uF	1-4
M2	PSFC, Vcc	51	3.0	5.6	uF	1-4
M3	PSFC, V2	33	0.9	1.7	uF	1-4
M4	PSFC, V3	50	1.5	2.7	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1-M4
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1-M4
Capacitance measurement	NA NA NA NA	± 60 nF ± 130 nF ± 40 nF ± 60 nF	M1 M2 M3 M4

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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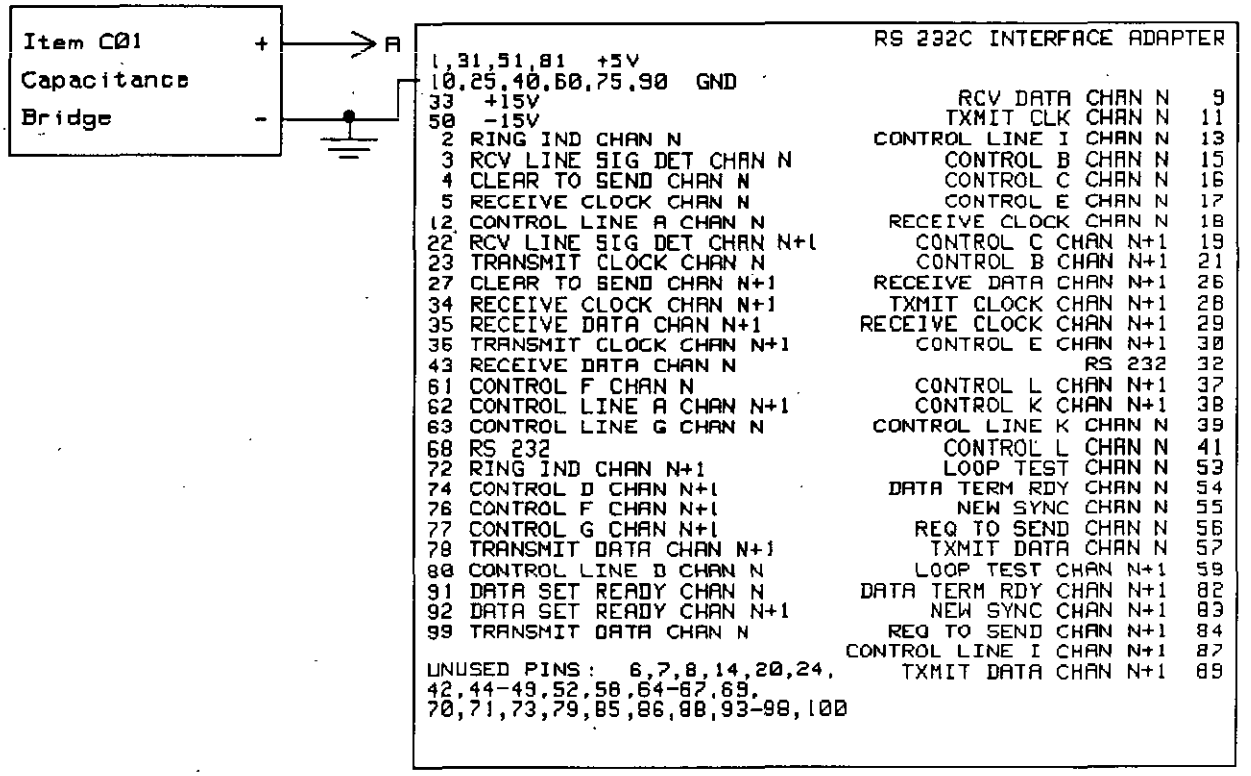


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	Icc, Vcc	1,31, 51,81	150.0	360.0	mA	1-3
N2	Icc, V2	33	17.0	41.0	mA	1-3
N3	Icc, V3	50	-9.4	-3.9	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1-N3
(V2)	16.5 V	± 165 mV	N1-N3
(V3)	-16.5 V	± 165 mV	N1-N3
Power supply current limit, (Vcc)	0.54 A	NA	N1-N3
(V2)	0.06 A	NA	N1-N3
(V3)	-0.02 A	NA	N1-N3
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1-N3
low, (VL)	0.0 V	± 100 mV	N1-N3
Power supply current measurement	NA	± 10.5 mA	N1
	NA	± 1.2 mA	N2
	NA	± 0.28 mA	N3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (10,000 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (10,000 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	P1
(V2)	15.0 V	± 150 mV	P1,P2
(V3)	-15.0 V	± 150 mV	P1,P2
(Vcc)	5.5 V	± 55 mV	P2
Power supply current limit, (Vcc)	0.54 A	NA	P1,P2
(V2)	0.06 A	NA	P1,P2
(V3)	-0.02 A	NA	P1,P2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) The pattern voltages shall be applied as follows:

Pin number	Pattern voltage				Test condition
	High (VH)	Accuracy	Low (VL)	Accuracy	
12,61,62,63,68, 74,76,77,78,80, 91,92,99	2.4 V	± 100 mV	0.4 V	± 100 mV	P1,P2
2,3,4,5,22,23, 27,34,35,36,43, 72	4.0 V	± 100 mV	-4.0 mV	± 100 mV	P1,P2

(c) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

Pin number	Output compare levels				Test condition
	High level	Accuracy	Low level	Accuracy	
9,11,13,15, 16,17,18,19, 21,26,28,29, 30,37,38,39, 41,87	2.4 V ≤ Voh ≤ Vcc	± 105 mV	0.0 V ≤ Vol ≤ 0.5 V	± 25 mV	P1
9,11,13,15, 16,17,18,19, 21,26,28,29, 30,37,38,39, 41,87	2.4 V ≤ Voh ≤ Vcc	± 155 mV	0.0 V ≤ Vol ≤ 0.5 V	± 25 mV	P2
53,54,55,56, 57,59,82,83, 84,89	5.0 V ≤ Voh ≤ Vcc 7.0 V	± 100 mV	-7.0 V ≤ Vol ≤ -5.0 V	± 100 mV	P1,P2

3. Abbreviations used are defined in 6.1.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (10,000 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (10,000 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (10,000 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (10,000 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Q1,Q3
(V2)	15.0 V	± 150 mV	Q1-Q4
(V3)	-15.0 V	± 150 mV	Q1-Q4
(Vcc)	5.5 V	± 55 mV	Q2,Q4
Power supply current limit, (Vcc)	0.54 A	NA	Q1-Q4
(V2)	0.06 A	NA	Q1-Q4
(V3)	-0.02 A	NA	Q1-Q4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The pattern voltages shall be applied as follows:

Pin number	Pattern voltage				Test condition
	High (VH)	Accuracy	Low (VL)	Accuracy	
12,61,62,63,68, 74,76,77,78,80, 91,92,99	2.4 V	± 100 mV	0.4 V	± 100 mV	Q1-Q4
2,3,4,5,22,23, 27,34,35,36,43, 72	4.0 V	± 100 mV	-4.0 V	± 100 mV	Q1-Q4

(c) The output compare levels shall be as follows:

Pin number	Output compare levels				Test condition
	High level	Accuracy	Low level	Accuracy	
9,11,13,15, 16,17,18,19, 21,26,28,29, 30,37,38,39, 41,87	Vref ≤ Voh ≤ Vcc	± 100 mV	0.0 V ≤ Vol ≤ Vref	± 100 mV	Q1,Q3
9,11,13,15, 16,17,18,19, 21,26,28,29, 30,37,38,39, 41,87	Vref ≤ Voh ≤ Vcc	± 100 mV	0.0 V ≤ Vol ≤ Vref	± 100 mV	Q2,Q4
53,54,55,56, 57,59,82,83, 84,89	Vref ≤ Voh ≤ 7.0 V	± 100 mV	-7.0 V ≤ Vol ≤ Vref	± 100 mV	Q1-Q4

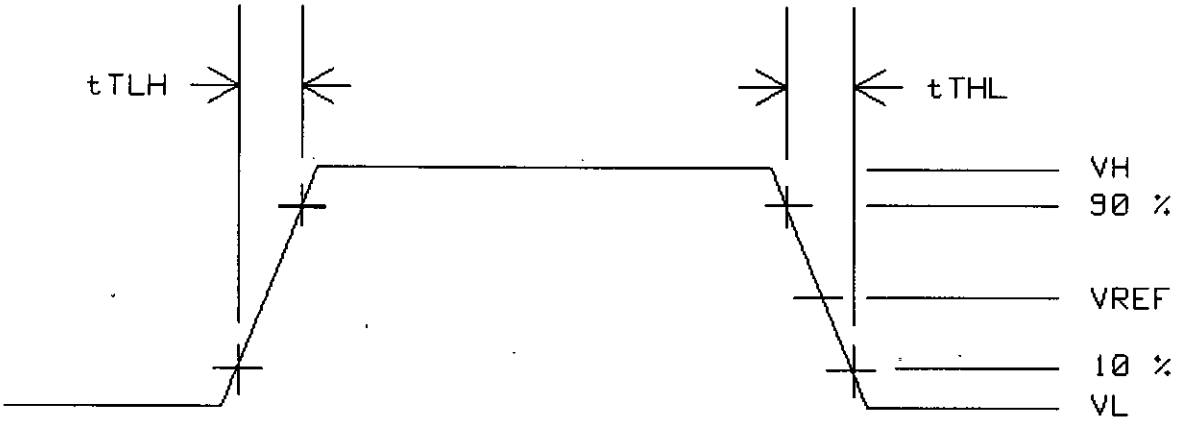
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NOTES: - Continued.

(d) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in note 2 (b).

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V for pins: 9,11,12,13,15,16,17,18,19,21,26, 28,29,30,37,38,39,41,61,62,63,68,74,76,77,78,80,87,91,92,99.

(d) Vref = 0.0 V for pins: 2,3,4,5,22,23,27,34,35,36,43,53, 54,55,56,57,59,72,82,83,84,89.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	RNG N to CTRL B N	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV LSD N to CTRL C N	-	-	-	3500	3500	3500	nS	11
1	tpd	CLR SND N to CTRL E N	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV CLK N to RCV CLK N	-	-	-	3500	3500	3500	nS	11
1	tpd	XMT CLK N to XMT CLK N	-	-	-	3500	3500	3500	nS	11
1	tpd	RNG N+1 to CTRL B N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV LSD N+1 to CTRL C N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	CLR SND N+1 to CTRL E N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	RCV CLK N+1 to RCV CLK N+1	-	-	-	3500	3500	3500	nS	11
1	tpd	XMT CLK N+1 to XMT CLK N+1	-	-	-	3500	3500	3500	nS	11
2	tpd	RCV D N to RCV D N	-	-	-	3527	3527	3527	nS	11
2	tpd	RCV D N+1 to RCV D N+1	-	-	-	3527	3527	3527	nS	11
3	tpd	D ST RDY N to CTRL K N	-	-	-	3545	3545	3545	nS	11

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	D ST RDY N+1 to CTRL K N+1	-	-	-	3545	3545	3545	nS	11
4	tpd	CTRL A N to LP TEST N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL D N to D TM RDY N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL F N to NEW SYNC N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL G N to RQ TO SD N	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL A N+1 to LP TEST N+1	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL D N+1 to D TM RDY N+1	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL F N+1 to NEW SYNC N+1	-	-	-	4500	4500	4500	nS	11
4	tpd	CTRL G N+1 to RQ TO SD N+1	-	-	-	4500	4500	4500	nS	11
5	tpd	XMT D N+1 to XMT D N+1	-	-	-	4527	4527	4527	nS	11
5	tpd	XMT D N to XMT D N	-	-	-	4527	4527	4572	nS	11

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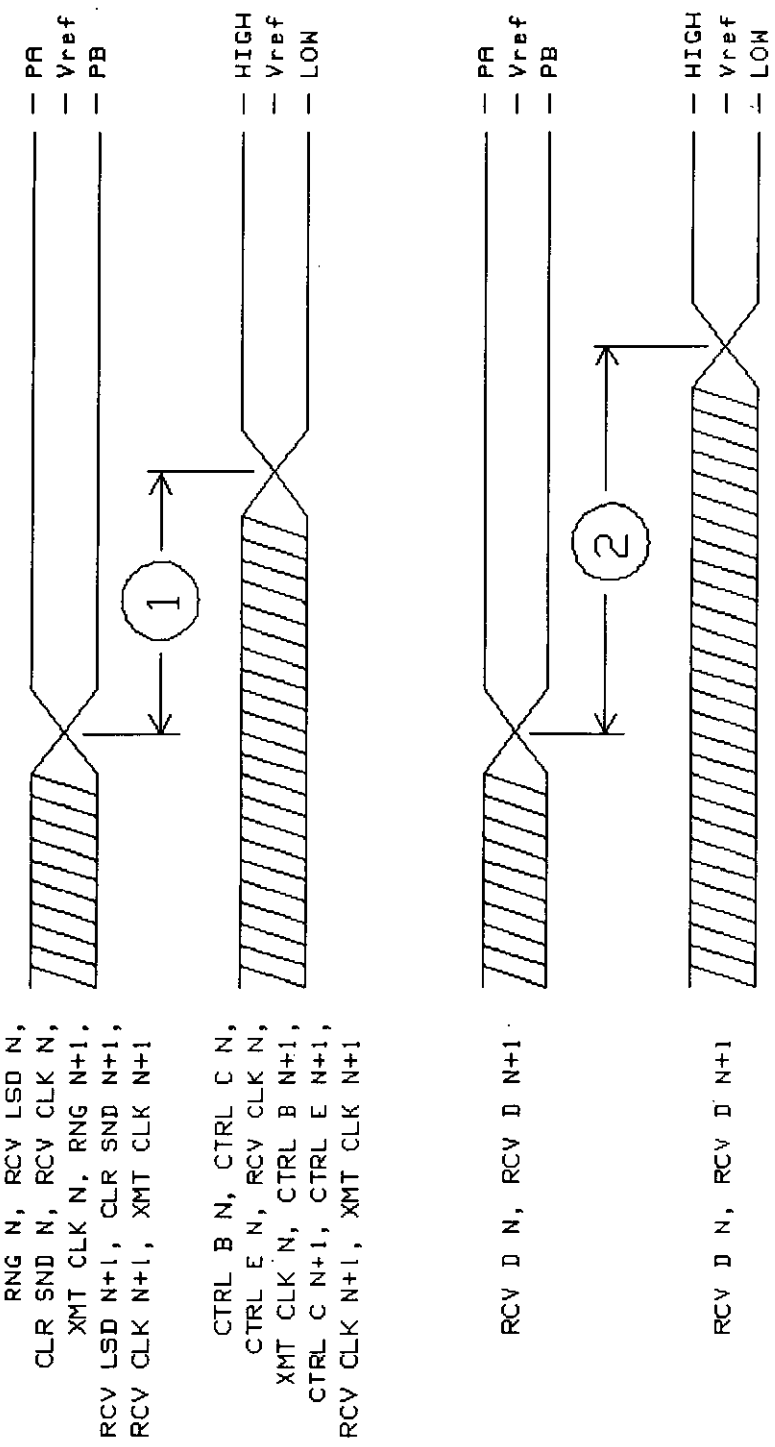


FIGURE 11a. Dynamic waveforms for table XVI.

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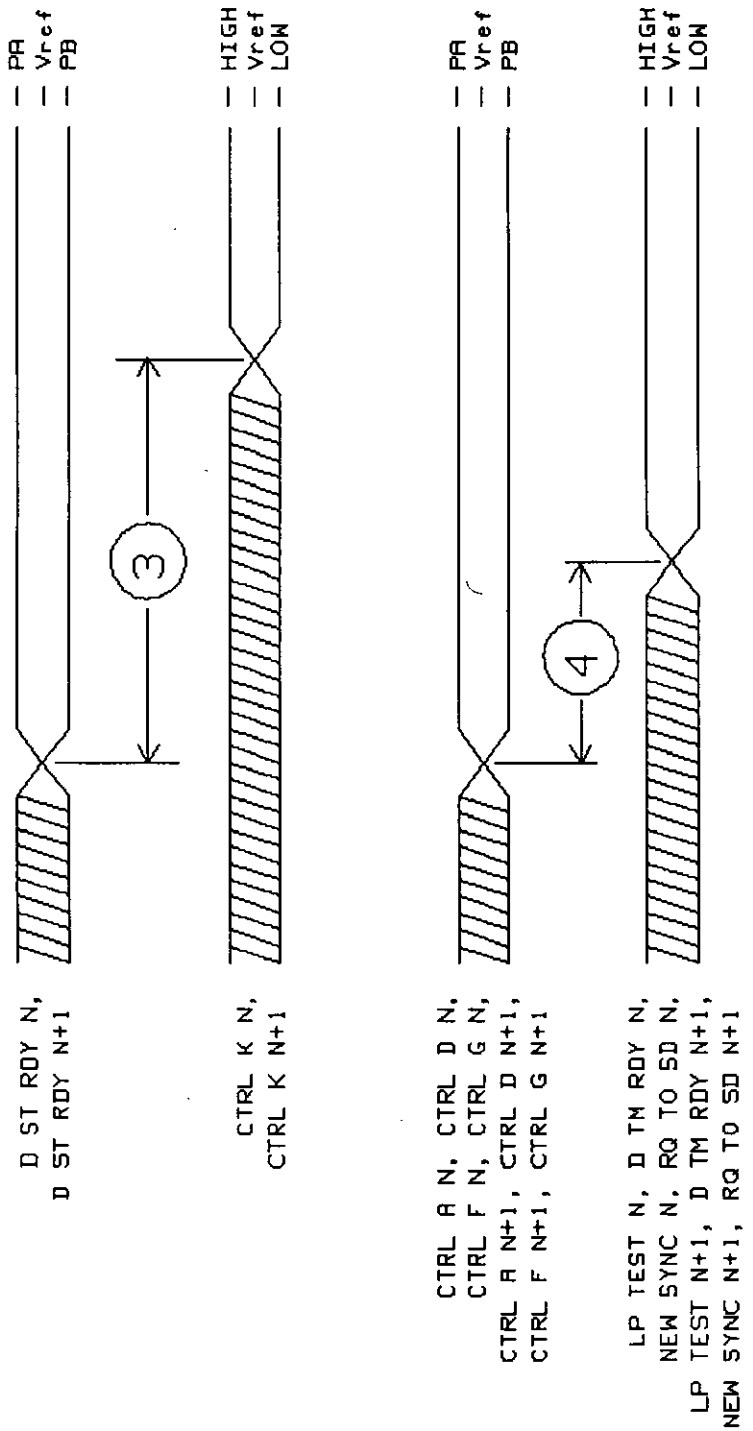


FIGURE 11b. Dynamic waveforms for table XVI.

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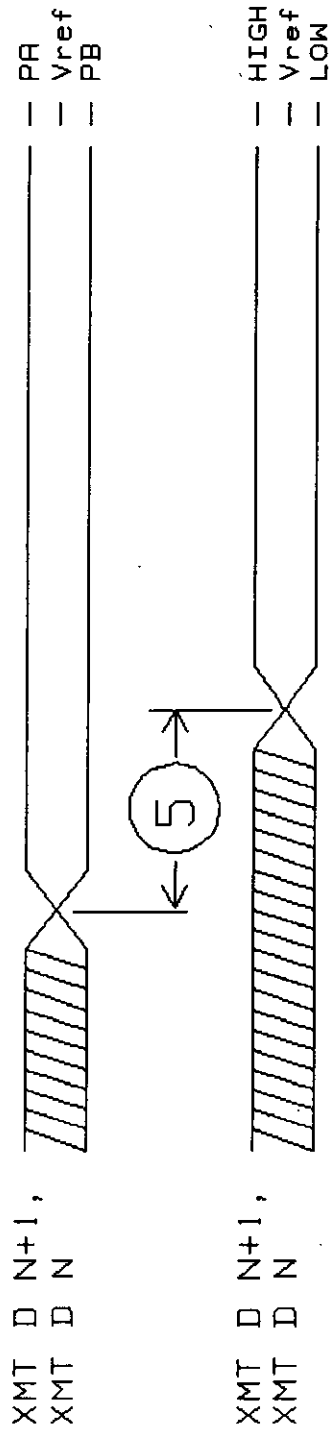


FIGURE 11c. Dynamic waveforms for table XVI.

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TABLE XVII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	RNG N	2	4	50	pF	1-4
R2	RCV LSD N	3				
R3	CLR SND N	4				
R4	RCV CLK N	5	4	50	pF	1-4
R5	CTRL A N	12	2	75	pF	1-4
R6	RCV LSD N+1	22	4	75	pF	1-4
R7	XMT CLK N	23	4	75	pF	1-4
R8	CLR SND N+1	27	4	50	pF	1-4
R9	RCV CLK N+1	34	4	75	pF	1-4
R10	RCV D N+1	35				
R11	XMT CLK N+1	36	4	75	pF	1-4
R12	RCV D N	43	4	100	pF	1-4
R13	CTRL F N	61	2	50	pF	1-4
R14	CTRL A N+1	62				
R15	CTRL G N	63	2	50	pF	1-4
R16	RNG N+1	72	4	75	pF	1-4
R17	CTRL D N+1	74	2	50	pF	1-4
R18	CTRL F N+1	76				
R19	CTRL G N+1	77	2	50	pF	1-4
R20	XMT D N+1	78	2	75	pF	1-4

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TABLE XVII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CTRL D N	80	2	75	pF	1-4
R22	D ST RDY N	91	4	50	pF	1-4
R23	D ST RDY N+1	92	4	50	pF	1-4
R24	XMT D N	99	2	75	pF	1-4

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R24
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R24
Capacitance measurement	NA	± 2.3 pF	R1-R4
	NA	± 3.65 pF	R5
	NA	± 3.55 pF	R6,R7
	NA	± 2.3 pF	R8
	NA	± 3.55 pF	R9-R11
	NA	± 4.8 pF	R12
	NA	± 2.4 pF	R13-R15
	NA	± 3.55 pF	R16
	NA	± 2.4 pF	R17-R19
	NA	± 3.65 pF	R20,R21
	NA	± 2.3 pF	R22,R23
	NA	± 3.65 pF	R24

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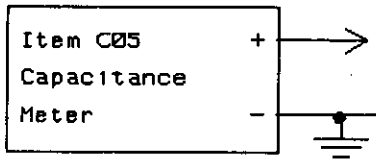
NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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RS 232C INTERFACE ADAPTER

1,31,51,81	+5V		
10,25,40,60,75,90	GND		
33	+15V	RCV DATA CHAN N	9
50	-15V	TXMIT CLK CHAN N	11
2	RING IND CHAN N	CONTROL LINE I CHAN N	13
3	RCV LINE SIG DET CHAN N	CONTROL B CHAN N	15
4	CLEAR TO SEND CHAN N	CONTROL C CHAN N	16
5	RECEIVE CLOCK CHAN N	CONTROL E CHAN N	17
12	CONTROL LINE A CHAN N	RECEIVE CLOCK CHAN N	18
22	RCV LINE SIG DET CHAN N+1	CONTROL C CHAN N+1	19
23	TRANSMIT CLOCK CHAN N	CONTROL B CHAN N+1	21
27	CLEAR TO SEND CHAN N+1	RECEIVE DATA CHAN N+1	26
34	RECEIVE CLOCK CHAN N+1	TXMIT CLOCK CHAN N+1	28
35	RECEIVE DATA CHAN N+1	RECEIVE CLOCK CHAN N+1	29
36	TRANSMIT CLOCK CHAN N+1	CONTROL E CHAN N+1	30
43	RECEIVE DATA CHAN N	RS 232	32
61	CONTROL F CHAN N	CONTROL L CHAN N+1	37
62	CONTROL LINE A CHAN N+1	CONTROL K CHAN N+1	38
63	CONTROL LINE G CHAN N	CONTROL LINE K CHAN N	39
68	RS 232	CONTROL L CHAN N	41
72	RING IND CHAN N+1	LOOP TEST CHAN N	53
74	CONTROL D CHAN N+1	DATA TERM RDY CHAN N	54
76	CONTROL F CHAN N+1	NEW SYNC CHAN N	55
77	CONTROL G CHAN N+1	REQ TO SEND CHAN N	56
78	TRANSMIT DATA CHAN N+1	TXMIT DATA CHAN N	57
80	CONTROL LINE D CHAN N	LOOP TEST CHAN N+1	59
91	DATA SET READY CHAN N	DATA TERM RDY CHAN N+1	82
92	DATA SET READY CHAN N+1	NEW SYNC CHAN N+1	83
99	TRANSMIT DATA CHAN N	REQ TO SEND CHAN N+1	84
		CONTROL LINE I CHAN N+1	87
UNUSED PINS:	6,7,8,14,20,24,	TXMIT DATA CHAN N+1	89
42,44-49,52,58,64-67,69,			
70,71,73,79,85,86,88,93-98,100			

FIGURE 12. Input capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T	PSTA, Vcc	1	0	400	mV	1-4
T	PSTA, Vcc	51	0	400	mV	1-4
T	PSTA, V2	33	0	400	mV	1-4
T	PSTA, V3	50	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1-T4
(V2)	15.0 V	± 750 mV	T1-T4
(V3)	-15.0 V	± 750 mV	T1-T4
Power supply current limit, (Vcc)	0.54 A	NA	T1-T4
(V2)	0.06 A	NA	T1-T4
(V3)	-0.02 A	NA	T1-T4
Pattern voltage high, (VH)	3.0 V	± 100 mV	T1-T4
low, (VL)	0.0 V	± 100 mV	T1-T4
Power supply transient amplitude measurement	NA	± 20 mV	T1-T4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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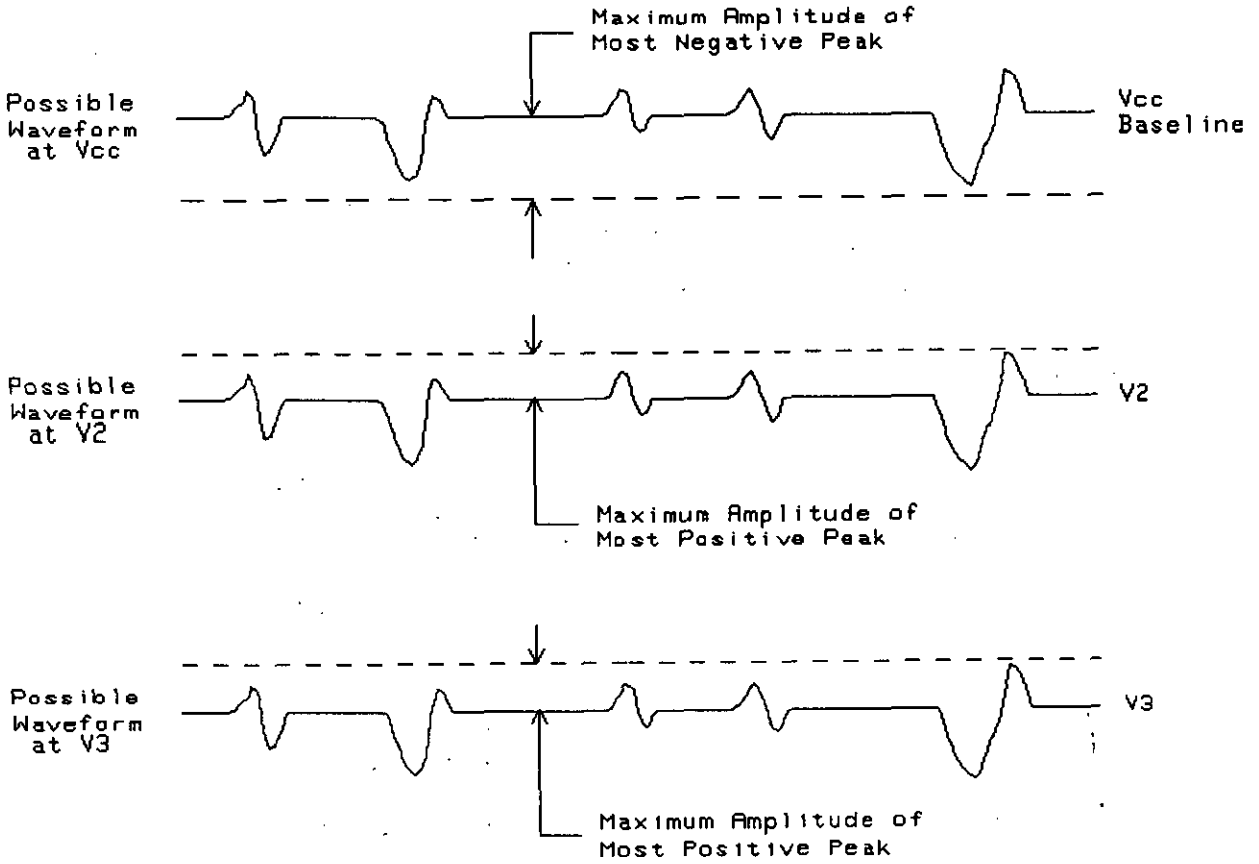
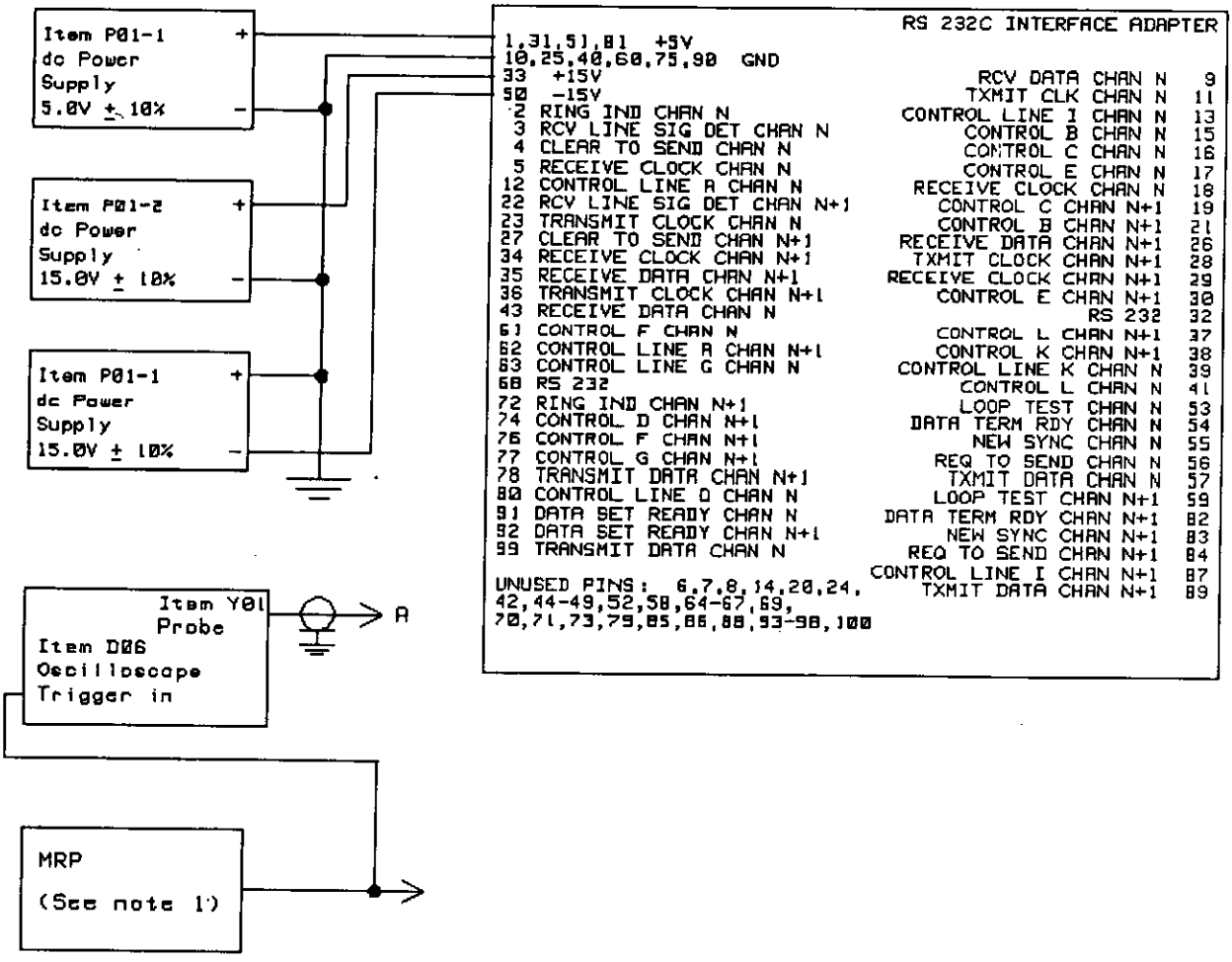


FIGURE 13. Power supply transient amplitude waveform for table XVIII.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XVIII.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/537	Modules, Standard Electronic, RS 232C Interface Adapter Module, Key Code AML

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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